

Optimization Methods for Machine Learning and Engineering

Lecture 8 – Duality

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Agenda

1. Lagrangian Duality
2. Karush-Kuhn-Tucker Conditions
3. Primal-Dual Methods

Why Duality?

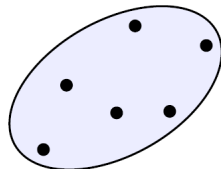
Many kinds of duality exist in mathematics

- **Two equivalent representations** of the same object with a **lossless conversion** between the two. Yields additional insight and/or algorithmic improvements.
- Example: Closed convex sets as the union of points or as the intersection of halfspaces
- Example: Fourier Transform of a time series

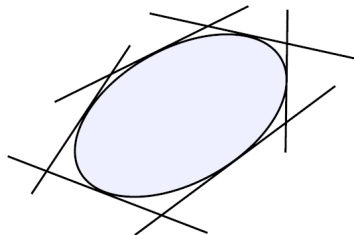
Why Duality in Optimization?

- Compute **lower bounds** for the optimum
- **Dimensionality reduction** and **faster convergence** by the consideration of **additional optimality conditions**

Sets as Unions of Points



Sets as Intersections of Halfspaces



Lagrangian Duality

von Neumann's Minimax Theorem

Let $\psi : X \times Y \rightarrow \mathbb{R}$. The **Minimax Inequality** guarantees that

$$\sup_{y \in Y} \inf_{x \in X} \psi(x, y) \leq \inf_{x \in X} \sup_{y \in Y} \psi(x, y).$$

Minimax Theorem

If X and Y are **compact** (closed and boundend) and **convex**, and $\psi(x, \cdot)$ is **convex in x** and $\psi(\cdot, y)$ is **concave in y** :

$$\max_{y \in Y} \min_{x \in X} \psi(x, y) = \min_{x \in X} \max_{y \in Y} \psi(x, y)$$

The Minimax Theorem can be thought of as a **two-player zero-sum game** situation [Neumann1928]. The gain of “Player X” is the loss of “Player Y” and vice versa. Both try to come up with an optimal play.

The infimum/supremum $\inf/\sup$ also considers the solution in the limit (e.g. on the closure of an open set or ∞). We use $\inf/\sup$ instead of $\min/\max$ when we are not on closed sets.



John von Neumann (1903 – 1957)

As far as I can see, there could be no theory of games [...] without that theorem. [...] I thought there was nothing worth publishing until the Minimax Theorem was proved.

Game Theory in a Nutshell

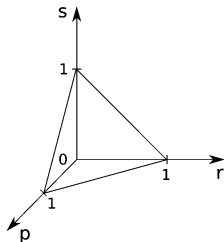
Rock-Paper-Scissors (Zero-Sum)

	Rock	Paper	Scissors
Rock	0	-1	1
Paper	1	0	-1
Scissors	-1	1	0

Prisoner's Dilemma (Not Zero-Sum)

	Talk	Don't Talk
Talk	(-5,-5)	(0, -10)
Don't Talk	(-10, 0)	(-2, -2)

Simplex for Mixed Strategies
(Rock-Paper-Scissors)



- Two-player zero-sum games are well-studied in Game Theory
- Many games are not zero-sum, e.g. the famous Prisoner's Dilemma
- A **pure strategy** is the choice of an option
- A **mixed strategy** assigns a probability to each pure strategy
 - For example $x = (0.1, 0.5, 0.4)$ in Rock-Paper-Scissors
 - How would you choose y if you were the other player?
- Finding the best strategy can be computationally very challenging. Humans are generally not capable of always playing the best strategy.

Existence of Saddle Point / Nash Equilibrium

A **Nash Equilibrium** is a “stable” combination of player strategies, where none of the players can improve his situation by changing his strategy unilaterally.

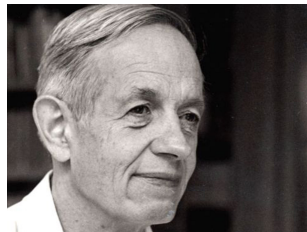
Nash showed that **at least one Nash Equilibrium must exist** in games where the set of mixed strategies is compact [Nash1950]. This is also the case for non-zero-sum games.

There can be many Nash Equilibria with different rewards. Often they are computationally hard to find [Shoham2008].

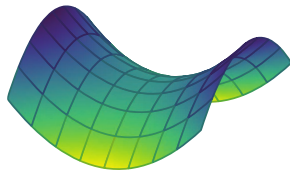
Existence of a Saddle Point

In the Minimax setting, there exists at least one **saddle point** $(x^*, y^*) \in X \times Y$ where

$$\min_{x \in X} \psi(x, y^*) = \psi(x^*, y^*) = \max_{y \in Y} \psi(x^*, y).$$



John Nash (1928 – 2015)



ψ has at least one saddle point

Minimax for Equality Constraints

$$\begin{aligned} & \inf_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}) \\ & \text{subject to } c_i(\mathbf{x}) = 0, \quad i = 1 \dots, m \end{aligned}$$

The shorthand for the constraints is $\mathbf{c}(\mathbf{x}) = \mathbf{0}$ with $\mathbf{c} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ with $X = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{c}(\mathbf{x}) = \mathbf{0}\}$ the **feasible solutions**. With the **indicator function** I_X we can state an **equivalent unconstrained problem**:

$$\tilde{f}(\mathbf{x}) = f(\mathbf{x}) + I_X(\mathbf{x}) \quad \Rightarrow \quad \inf_{\mathbf{x} \in X} f(\mathbf{x}) = \inf_{\mathbf{x} \in \mathbb{R}^n} \tilde{f}(\mathbf{x})$$

The essential observation is that we can **replace the indicator function with a supremum term**. The additional term $\sup_{\mathbf{y} \in \mathbb{R}^m} \mathbf{y}^\top (\mathbf{c}(\mathbf{x}))$ vanishes if $\mathbf{c}(\mathbf{x}) = \mathbf{0}$ and goes to infinity otherwise.

$$f(\mathbf{x}) + I_X(\mathbf{x}) = \sup_{\mathbf{y} \in \mathbb{R}^m} (f(\mathbf{x}) + \mathbf{y}^\top \mathbf{c}(\mathbf{x}))$$

The Minimax formulation of the constrained problem is

$$\inf_{\substack{\mathbf{x} \in \mathbb{R}^n \\ \mathbf{c}(\mathbf{x}) = \mathbf{0}}} f(\mathbf{x}) = \inf_{\mathbf{x} \in \mathbb{R}^n} \sup_{\mathbf{y} \in \mathbb{R}^m} f(\mathbf{x}) + \mathbf{y}^\top \mathbf{c}(\mathbf{x}).$$

Lagrangian Duality

Suppose now the problem is convex:

$$\inf_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x})$$

subject to $\mathbf{Ax} - \mathbf{b} = \mathbf{0}, \quad \mathbf{A} \in \mathbb{R}^{m \times n}$

Minimax Formulation

$$\inf_{\mathbf{x} \in \mathbb{R}^n} \sup_{\mathbf{y} \in \mathbb{R}^m} f(\mathbf{x}) + \mathbf{y}^\top (\mathbf{Ax} - \mathbf{b})$$

Lagrangian Formulation

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \boldsymbol{\lambda}^\top (\mathbf{Ax} - \mathbf{b})$$
$$\nabla \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = \mathbf{0}$$

Both approaches try to find the (only) saddle-point for the same equation. But for different motivations. The Minimax formulation **exposes additional problem structure** that we can exploit.

Primal Problem

$$\inf_{\mathbf{x} \in \mathbb{R}^n} \sup_{\boldsymbol{\lambda} \in \mathbb{R}^m} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda})$$

Dual Problem

$$\sup_{\boldsymbol{\lambda} \in \mathbb{R}^m} \inf_{\mathbf{x} \in \mathbb{R}^n} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda})$$

Reversing the inf/sup operations yields the **Lagrangian Dual:** $q(\boldsymbol{\lambda}) = \inf_{\mathbf{x} \in \mathbb{R}^n} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda})$

Weak/Strong Duality and the Duality Gap

From the Minimax Inequality follows **weak duality**:

$$\sup_{\lambda \in \mathbb{R}^m} q(\lambda) = \sup_{\lambda \in \mathbb{R}^m} \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \lambda) \leq \inf_{x \in \mathbb{R}^n} \sup_{\lambda \in \mathbb{R}^m} \mathcal{L}(x, \lambda)$$

The difference is the called the **duality gap**:

$$\inf_{x \in \mathbb{R}^n} \sup_{\lambda \in \mathbb{R}^m} \mathcal{L}(x, \lambda) - \sup_{\lambda \in \mathbb{R}^m} \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \lambda)$$

If the **duality gap is zero for** (x^*, λ^*) we have **strong duality**. In the case of strong duality, we know that in the optimum the dual has the same value as the original problem: $\sup_{\lambda \in \mathbb{R}^m} q(\lambda) = f(x^*)$

This is easy to show, as $\lambda^\top (\mathbf{A}x^* - \mathbf{b}) = 0$ vanishes in the Lagrangian.

The Minimax Theorem tells us that strong duality is present if $\mathcal{L}$ is convex in x and concave in λ , and if we have convexity and compactness in the function domains. If we don't have compactness, additional conditions can be checked to show strong duality.

Slater's Conditions [Slater1959]

There are several conditions to show strong duality, called **constraint qualifications**.

The most used constraint qualification is via **Slater's Conditions**:

Slater's Theorem

Let $f : X \rightarrow \mathbb{R}$ a convex optimization problem over the (convex) primal feasible set X . If there **exists a solution x in the relative interior of X** , then f has **strong duality**.

Let X the set of points. The **relative interior** is defined as the interior with respect to its **affine hull**, all affine combinations made from elements of X . Hence, if equality constraints $\mathbf{A}\mathbf{x} - \mathbf{b} = \mathbf{0}$ have “flattened out” the feasible set within the embedding space $\mathbb{R}^n$, then there can still be solutions in the relative interior.

$$\text{aff}(X) = \left\{ \sum_{i=1}^k \alpha_i \mathbf{x}_i \mid k > 0, \mathbf{x}_i \in X, \alpha_i \in \mathbb{R}, \sum_{i=1}^k \alpha_i = 1 \right\}$$
$$\text{relint}(X) = \{ \mathbf{x} \in X : \exists \varepsilon > 0, B(\mathbf{x}, \varepsilon) \cap \text{aff}(X) \subseteq X \}$$

With this, we can formally write Slater's Condition as $\text{relint}(X) \neq \emptyset$.

Slater's Theorem also holds when the optimization problem contains inequality constraints.

We will see the definition for a dual that includes inequality constraints later on.

Closed-Form Solutions with the Lagrangian Dual

Consider the quadratic convex program $\inf_{\mathbf{x} \in \mathbb{R}^n} \frac{1}{2} \mathbf{x}^\top \mathbf{x}$, subject to $\mathbf{A}\mathbf{x} - \mathbf{b} = \mathbf{0}$, $\mathbf{A} \in \mathbb{R}^{m \times n}$.

With $\sup_{\lambda \in \mathbb{R}^m} \inf_{\mathbf{x} \in \mathbb{R}^n} \mathcal{L}(\mathbf{x}, \lambda) = \sup_{\lambda \in \mathbb{R}^m} q(\lambda)$ the Lagrangian dual is $q(\lambda) = \inf_{\mathbf{x} \in \mathbb{R}^n} \frac{1}{2} \mathbf{x}^\top \mathbf{x} + \lambda^\top (\mathbf{A}\mathbf{x} - \mathbf{b})$.

For the infimum term we know the **closed form solution** by setting the gradient to zero:

$$\mathbf{x} + \mathbf{A}^\top \lambda = \mathbf{0} \quad \Rightarrow \quad \mathbf{x} = -\mathbf{A}^\top \lambda \quad (1)$$

For the supremum term, plug in the result to get

$$\sup_{\lambda \in \mathbb{R}^m} \frac{1}{2} \underbrace{(-\mathbf{A}^\top \lambda)^\top}_{\mathbf{x}} \underbrace{(-\mathbf{A}^\top \lambda)}_{\mathbf{x}} + \lambda^\top (\mathbf{A} \underbrace{(-\mathbf{A}^\top \lambda)}_{\mathbf{x}} - \mathbf{b}) = \sup_{\lambda \in \mathbb{R}^m} -\frac{1}{2} \lambda^\top (\mathbf{A}\mathbf{A}^\top) \lambda - \lambda^\top \mathbf{b}.$$

We know that $\mathbf{A}\mathbf{A}^\top = (\mathbf{A}^\top)^\top \mathbf{A}^\top = (\mathbf{A}\mathbf{A}^\top)^\top$ is symmetric.

It is also positive definite: $\lambda^\top \mathbf{A}\mathbf{A}^\top \lambda = (\mathbf{A}^\top \lambda)^\top (\mathbf{A}^\top \lambda) = \|\mathbf{A}^\top \lambda\|_2^2 \geq 0$.

Set the gradient of the supremum term to zero for the **closed form solution** $\lambda^* = (\mathbf{A}\mathbf{A}^\top)^{-1} \mathbf{b}$.

We have **strong duality**. So together with (1), the primal optimum is $\mathbf{x}^* = -\mathbf{A}^\top (\mathbf{A}\mathbf{A}^\top)^{-1} \mathbf{b}$.

Lower Bounds for a Non-Convex Problem

Consider the problem $\inf_{\mathbf{x} \in \mathbb{R}^n} \mathbf{x}^\top \mathbf{Q} \mathbf{x}$, subject to $x_i^2 = 1, \quad i = 1, \dots, n$ with $\mathbf{Q}$ positive definite.

The problem is **non-convex** because the **primal-feasible set contains discrete solutions** with $x_i \in \{1, -1\}$.

- The size of the feasible set is 2^n , **growing exponentially**.
- The **Slater Conditions don't hold for this problem**, so we only have weak duality.

The Lagrangian dual is $q(\boldsymbol{\lambda}) = \inf_{\mathbf{x} \in \mathbb{R}^n} \mathbf{x}^\top \mathbf{Q} \mathbf{x} + \sum_{i=1}^n \lambda_i (x_i^2 - 1) = \inf_{\mathbf{x} \in \mathbb{R}^n} \mathbf{x}^\top \underbrace{(\mathbf{Q} + \text{diag}(\boldsymbol{\lambda}))}_{\mathbf{V}} \mathbf{x} - \mathbf{1}^\top \boldsymbol{\lambda}$.

Note that $q(\boldsymbol{\lambda})$ tends to $-\infty$ if $\mathbf{V}$ is not positive semi-definite. But as we know that $q(\mathbf{0}) = 0$, **all solutions $\boldsymbol{\lambda}$ for which $\mathbf{V}$ is not positive semi-definite can be excluded**.

A good lower bound from the literature: Choose $\bar{\boldsymbol{\lambda}} = -\mathbf{1} \cdot \text{eig}_{\min}(\mathbf{Q})$ with $\text{eig}_{\min}(\mathbf{Q})$ the smallest Eigenvalue of $\mathbf{Q}$; a small $\bar{\boldsymbol{\lambda}}$ for which we still know $\mathbf{V}$ to be positive semi-definite.

Even though we could not find the optimum solution via the Lagrangian Dual, we have $\sup_{\boldsymbol{\lambda} \in \mathbb{R}^m} q(\boldsymbol{\lambda}) > q(\bar{\boldsymbol{\lambda}}) = -\mathbf{1}^\top \bar{\boldsymbol{\lambda}} = n \cdot \text{eig}_{\min}(\mathbf{Q})$ as a **lower bound for the original problem**.

Karush-Kuhn-Tucker Conditions

Lagrangian with Inequality Constraints

Recall the canonical form of constrained optimization problems with **inequality** and **equality** constraints.

$$\begin{array}{ll} \min_{\mathbf{x} \in \mathbb{R}^n} & f(\mathbf{x}) \\ \text{subject to} & \mathbf{g}_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, m \\ & \mathbf{h}_j(\mathbf{x}) = 0, \quad j = 1, \dots, l \end{array}$$

- Since we only consider convex problems, the equality conditions have to be affine with $\mathbf{h}_j(\mathbf{x}) = \mathbf{a}_j^\top \mathbf{x} - b_j$.
- The constraints combined in a single expression:
 - Inequality Constraints: $\mathbf{g}(\mathbf{x}) \leq \mathbf{0}$ for $\mathbf{g}(\mathbf{x}) = \begin{pmatrix} g_1(\mathbf{x}) \\ \vdots \\ g_m(\mathbf{x}) \end{pmatrix}$
 - Equality Constraints: $\mathbf{Ax} - \mathbf{b} = \mathbf{0}$
- Equality constraints can be incorporated in a Lagrangian

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \boldsymbol{\lambda}^\top (\mathbf{Ax} - \mathbf{b}) \quad \text{where} \quad \nabla \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*) = \mathbf{0}$$

Revisiting the Lagrangian with Added Log-Barrier

The Lagrangian for equality constraints with the added Log-Barrier for the inequality constraints is $\tilde{\mathcal{L}}_t(\mathbf{x}, \boldsymbol{\lambda})$. For $t \rightarrow \infty$ the optimizer is found as $\nabla \tilde{\mathcal{L}}_t(\mathbf{x}, \boldsymbol{\lambda}) = \mathbf{0}$.

$$\begin{aligned}\tilde{\mathcal{L}}_t(\mathbf{x}, \boldsymbol{\lambda}) &= f(\mathbf{x}) + \sum_{i=1}^m \left[-\frac{1}{t} \log(-g_i(\mathbf{x})) \right] + \boldsymbol{\lambda}^\top (\mathbf{A}\mathbf{x} - \mathbf{b}) \\ \nabla \tilde{\mathcal{L}}_{\mathbf{x}}(\mathbf{x}, \boldsymbol{\lambda}) &= \nabla f(\mathbf{x}) + \sum_{i=1}^m \left[\overbrace{\frac{-1}{t g_i(\mathbf{x})}}^{\text{Thought-Experiment: } \mu_i} \nabla g_i(\mathbf{x}) \right] + \mathbf{A}^\top \boldsymbol{\lambda}\end{aligned}$$

What happens with the “though-experiment” if $t \rightarrow \infty$? What if $g_i(\mathbf{x}) < 0$ or $g_i(\mathbf{x}) \rightarrow 0$?

We now introduce an **additional Lagrange-multiplier μ** for the inequality constraints:

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\mu}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \boldsymbol{\mu}^\top \mathbf{g}(\mathbf{x}) + \boldsymbol{\lambda}^\top (\mathbf{A}\mathbf{x} - \mathbf{b})$$

Imagine the case $\mu_i = \frac{-1}{t g_i(\mathbf{x})}$. This would exactly recover the Lagrangian with the added Log-Barrier. However the μ_i are now a scalar that no longer depends on $\mathbf{x}$.

Let us **use this additional freedom to select μ_i** . But with the **constraint that $\mu_i \geq 0$** because $g_i(\mathbf{x}) \leq 0$ (otherwise we end up on the wrong side of the log-barrier in the “though-experiment”).

Karush-Kuhn-Tucker Optimality Conditions

Deriving this extended Lagrangian by μ gives $\nabla_{\mu} \mathcal{L}(x, \mu, \lambda) = g(x)$. But we cannot require $g(x^*) = 0$!

Instead, if the optimizer does not lie on the “border” of the inequality constraint, we would prefer not to see that constraint at all. This is expressed by the **complementary slackness** $\mu_i g_i(x^*) = 0$. For all i either

- $g_i(x^*) = 0$ and we treat the Lagrange multiplier μ_i as if it was an equality constraint.
- $g_i(x^*) < 0$ and we require $\mu_i = 0$ to “shut off” the constraint.

If the Slater Conditions hold, the **KKT Conditions are sufficient to show optimality** of a solution [Kuhn1951]. It was later discovered, that the same had already been proved in the unpublished master’s thesis [Karush39].

The KKT Optimality Conditions

$$\nabla f(x) + [\mathcal{J}g(x)]^{\top} \mu + \mathbf{A}^{\top} \lambda = 0$$

$$\mathbf{A}x - b = 0$$

$$g_i(x) \leq 0, \quad \mu_i \geq 0, \quad \mu_i g_i(x) = 0, \quad i = 1, \dots, m$$

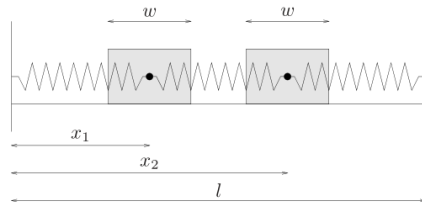
The so-called **Jacobian** of the inequality constraints is the matrix

$$\mathcal{J}g(x) = \begin{pmatrix} \nabla g_1(x)^{\top} \\ \vdots \\ \nabla g_m(x)^{\top} \end{pmatrix}.$$

The KKT Conditions describe properties of the optimizer, not a precise update step. We can use them with different algorithms (step directions), as long as they converge to a solution holding the KKT Conditions.

Mechanical Interpretation of the KKT Conditions [Boyd04]

- Left wall is at position zero, right wall at position l
- Blocks have width w
- x_1, x_2 are the displacements of blocks 1 and 2
- Springs with strength k_i between blocks/walls



The potential energy in the springs is a function of the block positions. The derivative of the potential energy is the force exerted by the springs. The system is in equilibrium if the potential energy is minimal.

$$f(x_1, x_2) = \frac{1}{2} (k_1 x_1^2 + k_2 (x_2 - x_1)^2 + k_3 (l - x_2)^2)$$

The equilibrium state is subject to kinematic constraints: Blocks cannot penetrate each other or the walls

$$g_1(x_1) = \frac{w}{2} - x_1 \leq 0, \quad g_2(x_1, x_2) = w + x_1 - x_2 \leq 0, \quad g_3(x_2) = \frac{w}{2} - l + x_2 \leq 0$$

When represented in the KKT Conditions, if blocks/walls come in contact, it is as if the “spring force” between them was increased so much that they will no longer overlap.

Primal-Dual Methods

Primal-Dual Methods

In a single step, Primal-Dual Methods **update both the primal variable x and the dual variables (μ, λ)** , in order to **eventually satisfy the KKT conditions**. The intermediary steps are not necessarily dual feasible.

Most Primal-Dual Methods are interior-point algorithms. A barrier is used to ensure solutions are always primal feasible and increasingly tightened. The update ensures that $\mu > 0$, also for the intermediary steps.

There are many variations. We focus on the Primal-Dual Method from [Boyd04].

It solves for a **modified KKT system**:

$$\begin{aligned}\nabla f(x) + [\mathcal{J}g(x)]^\top \mu + \mathbf{A}^\top \lambda &= \mathbf{0} \\ \mathbf{A}x - \mathbf{b} &= \mathbf{0} \\ g_i(x_i) \leq 0, \quad \mu_i \geq 0, \quad \mu_i g_i(x) + 1/t &= 0, \quad i = 1, \dots, m\end{aligned}$$

The “modification” is the additional term $1/t$ for the inequality constraints.

We increase t over the iterations of the Primal-Dual algorithm.

The original KKT conditions are approximated for large t .

The Newton Step of the Primal-Dual Method

The **residual** for the modified KKT system is $\mathbf{r}_t(\mathbf{x}, \boldsymbol{\mu}, \boldsymbol{\lambda}) = \begin{pmatrix} \nabla f(\mathbf{x}) + [\mathcal{J}\mathbf{g}(\mathbf{x})]^\top \boldsymbol{\mu} + \mathbf{A}^\top \boldsymbol{\lambda} \\ -\text{diag}(\boldsymbol{\mu})\mathbf{g}(\mathbf{x}) - \frac{1}{t}\mathbf{1} \\ \mathbf{Ax} - \mathbf{b} \end{pmatrix} = \mathbf{0}$.

We write it shorter as $\mathbf{r}_t(\mathbf{z})$, as a function of a single vector argument $\mathbf{z} = (\mathbf{x}, \boldsymbol{\mu}, \boldsymbol{\lambda})^\top$.

The update step $\Delta\mathbf{z}$ is computed with the Newton Method as $\mathcal{J}\mathbf{r}_t(\mathbf{z})\Delta\mathbf{z} = -\mathbf{r}_t(\mathbf{z})$:

$$\begin{pmatrix} \nabla^2 f(\mathbf{x}) + \sum_{i=1}^m \mu_i \nabla^2 g_i(\mathbf{x}) & \mathcal{J}\mathbf{g}(\mathbf{x})^\top & \mathbf{A}^\top \\ -\text{diag}(\boldsymbol{\mu})\mathcal{J}\mathbf{g}(\mathbf{x}) & -\text{diag}(\mathbf{g}(\mathbf{x})) & \mathbf{0} \\ \mathbf{A} & \mathbf{0} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \Delta\mathbf{x} \\ \Delta\boldsymbol{\mu} \\ \Delta\boldsymbol{\lambda} \end{pmatrix} = - \begin{pmatrix} \mathbf{r}_{\text{dual},t}(\mathbf{z}) \\ \mathbf{r}_{\text{cent},t}(\mathbf{z}) \\ \mathbf{r}_{\text{pri},t}(\mathbf{z}) \end{pmatrix}$$

The equation can be further shrunk by solving the second line for $\Delta\boldsymbol{\mu}$ and plugging the result into the first line (the second line can be removed entirely). See [Boyd04] for details.

Having the Newton step in place, we are left with two open question:

1. How to tighten the barrier by increasing t ?
2. How to ensure that we always have $\boldsymbol{\mu} \geq \mathbf{0}$ after the update step?

Basic Primal-Dual Interior-Point Method

LINE 4: We would use the Duality Gap to evaluate the distance to the optimum. But the intermediate steps are not necessarily feasible. Instead use the

Surrogate Duality Gap $\eta = \mathbf{g}(\mathbf{x})^\top \boldsymbol{\mu}$, which would be the Duality Gap if the solution was feasible with $\mathbf{r}_{\text{pri},t}(\mathbf{z}) = \mathbf{0}$ and $\mathbf{r}_{\text{dual},t}(\mathbf{z}) = \mathbf{0}$.

LINE 5: **Tighten the barrier** $t = \rho \frac{m}{\eta}$ as the Surrogate Duality Gap shrinks. A typical value for the scaling factor is $\rho = 10$.

LINE 6: **Compute the step direction** via the Newton Method to approach $\mathbf{r}_t(\mathbf{z}) = \mathbf{0}$.

LINE 7: **Modified backtracking linesearch** ensures that $\boldsymbol{\mu} \geq \mathbf{0}$ and $\mathbf{g}(\mathbf{x}) < \mathbf{0}$. A typical value for the step reduction is $p = 0.5$ and $\beta = 0.01$ for the minimum descent.

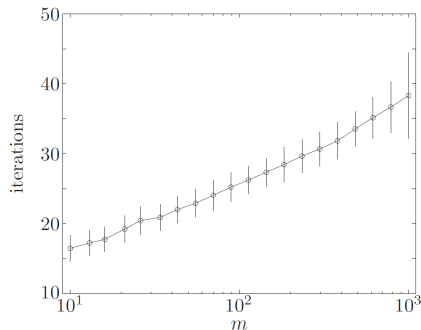
```
1: procedure PRIMALDUAL( $f, \mathbf{x}_0, \mathbf{g}, \mathbf{A}, \mathbf{b}, \varepsilon, \varepsilon_{\text{feas}}, \rho, p, \beta$ )
2:    $\mathbf{z} \leftarrow (\mathbf{x}_0, \boldsymbol{\mu}, \boldsymbol{\lambda})$  with initial  $\boldsymbol{\mu} > \mathbf{0}$ 
3:   repeat
4:      $\eta \leftarrow -\mathbf{g}(\mathbf{x})^\top \boldsymbol{\mu}$ 
5:      $t \leftarrow \rho \frac{m}{\eta}$ 
6:      $\Delta \mathbf{z} \leftarrow -[\mathcal{J} \mathbf{r}_t(\mathbf{z})]^{-1} \mathbf{r}_t(\mathbf{z})$ 
7:      $\alpha \leftarrow \text{PD-LINESEARCH}(\mathbf{r}_t, \mathbf{g}, \mathbf{z}, \Delta \mathbf{z}, p, \beta)$ 
8:      $\mathbf{z} \leftarrow \mathbf{z} + \alpha \Delta \mathbf{z}$ 
9:   until  $\|\mathbf{r}_{\text{dual},t}(\mathbf{z})\|_2 \leq \varepsilon_{\text{feas}} \wedge$   

    $\|\mathbf{r}_{\text{pri},t}(\mathbf{z})\|_2 \leq \varepsilon_{\text{feas}} \wedge \eta \leq \varepsilon$ 
10:  return  $\mathbf{z}_x$ 
11: end procedure

1: procedure PD-LINESEARCH( $\mathbf{r}_t, \mathbf{g}, \mathbf{z}, \Delta \mathbf{z}, p, \beta$ )
2:    $\alpha_{\text{max}} \leftarrow \min(\{1\} \cup \{i: \Delta z_{\mu_i} < 0\} \{-z_{\mu_i} / \Delta z_{\mu_i}\})$ 
3:    $\alpha \leftarrow 0.99 \alpha_{\text{max}}$ 
4:   while  $\exists i, g_i(\mathbf{z}_x + \Delta \mathbf{z}_x) \geq 0 \vee$   

      $\|\mathbf{r}_t(\mathbf{z} + \alpha \Delta \mathbf{z})\|_2 > (1 - \alpha \beta) \|\mathbf{r}_t(\mathbf{z})\|_2$  do
5:      $\alpha \leftarrow \alpha p$ 
6:   end while
7:   return  $\alpha$ 
8: end procedure
```

Convergence Speed of Primal-Dual Methods [Boyd04]



Number of iterations required to solve randomly generated standard LPs of different dimensions, with $n = 2m$. Error bars show standard deviation, around the average value, for 100 instances of each dimension. The growth in the number of iterations required, as the problem dimensions range over a 100:1 ratio, is approximately logarithmic.

Summary of what you learned today

- **The Minimax Theorem** and its importance in Game Theory and beyond
- **Lagrangian Duality** a way to “rewrite” an optimization problem that gives the same result (if it is convex)
 - We saw how the Lagrangian dual lets us simplify/solve/lower-bound example problems in ways that were not possible before
- We introduced a **Lagrange-Multiplier μ for the inequality constraints**
- **The KKT Conditions** need to be satisfied by the optimizer of an optimization problem
 - The KKT Conditions need to hold eventually. Intermediary steps of iterative algorithms can be primal/dual infeasible.
- **Primal-Dual Methods** are optimization algorithms that **update the primal and dual variables (Lagrange Multipliers) in a single step**

That's it for today.

See you next week for Lecture 9 on
Kernels and the Support Vector Machine

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